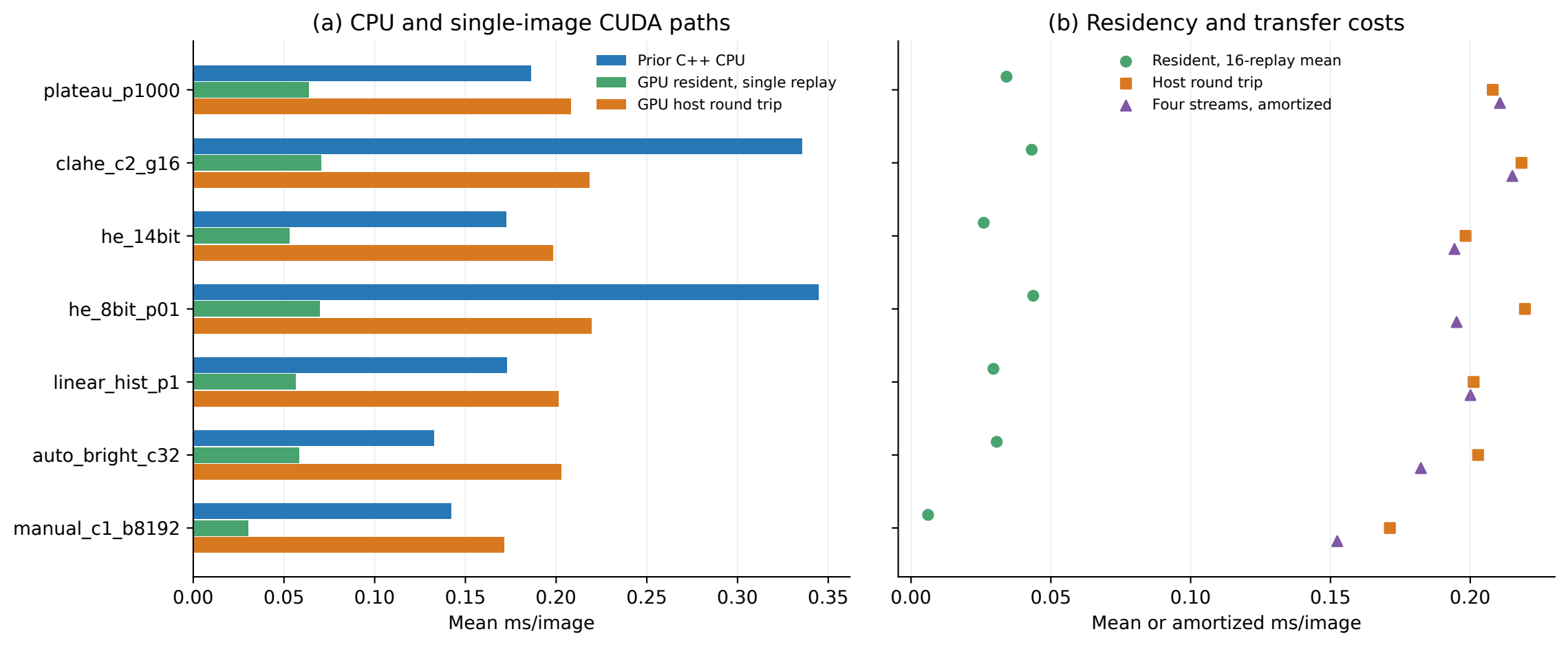
(a) CPU and single-image CUDA paths
Prior C++ CPU
GPU resident, single replay
GPU host round trip
plateau_p1000
clahe_c2_g16
he_14bit
he_8bit_p01
linear_hist_p1
auto_bright_c32
manual_c1_b8192
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
Mean ms/image
(b) Residency and transfer costs
Resident, 16-replay mean
Host round trip
Four streams, amortized
0.00
0.05
0.10
0.15
0.20
Mean or amortized ms/image